

BRAIN TUMOR CLASSIFICATION



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Introduction

Brain tumor classification is a critical task in medical imaging that plays a vital role in early diagnosis and effective treatment planning. MRI scans are widely used for detecting brain abnormalities; however, manual analysis by radiologists can be time-consuming and prone to variability. With the advancement of deep learning, automated systems have been developed to improve accuracy and efficiency in tumor classification. This project focuses on developing a robust brain tumor classification system using advanced models such as EfficientNetB0, DenseNet121, and ProtoPNet. In addition to classification, the system incorporates tumor localization using image processing techniques to provide visual insights into affected regions. The proposed approach aims to enhance diagnostic accuracy, improve interpretability, and assist medical professionals in making faster and more reliable decisions.

Literature Review

[1] S. Majdi, "Brain tumor classification using DenseNet and EfficientNet: A comparative study," *Comput. Methods Programs Biomed.*, vol. 237, p. 107567, 2024.

[2] M. J. Islam and Y. Zhang, "Brain tumor classification using MRI images and deep learning techniques," *PLOS ONE*, vol. 19, no. 5, p. e0301234, 2024.

[3] A. Iqbal et al., "Enhancing brain tumour multi-classification using EfficientNet-B0 architecture," *Information*, vol. 15, no. 8, 2024.

[4] Y. Chen, J. Li, and C.-J. Hsieh, "This looks like that: Deep learning with prototypes," in *Proc. NeurIPS*, 2019, pp. 13963–13974.

[5] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," *arXiv preprint arXiv:1905.11946*, 2019.

Research Gaps

1. Most systems focus only on classification and do not combine localization and interpretability in a single framework.
2. Models act as black boxes and do not provide visual insights into tumor regions.
3. Models struggle to adapt to different MRI datasets and require high computational resources.

Project Objectives

- 1.To develop an accurate deep learning model for multi-class brain tumor classification using MRI images.
- 2.To integrate multiple models (EfficientNetB0, DenseNet121, ProtoPNet) to improve performance and reliability.
- 3.To provide tumor localization and interpretability using image processing and explainable AI techniques.

Proposed Methodology

The proposed methodology follows a structured pipeline for brain tumor classification and localization using MRI images. The input images are fed into advanced deep learning models such as EfficientNetB0 and DenseNet121 for feature extraction and classification using transfer learning, while ProtoPNet is used to provide interpretability of predictions. A voting-based ensemble approach is applied to combine the outputs of all models and improve overall accuracy and reliability. Finally, tumor localization is performed using OpenCV-based image processing techniques, including thresholding and contour detection, to identify and highlight tumor regions within the MRI images.

Results and Discussion

INDIVIDUAL MODEL PREDICTION

EfficientNet: Glioma (65.21%)
DenseNet: Meningioma (98.92%)
ProtoPNet: Meningioma (89.26%)

FINAL ENSEMBLE RESULT

Prediction: Meningioma
Confidence: 98.92%
Model Used: DenseNet

TUMOR LOCALIZATION →

Testing & Validation

1. During training, EfficientNetB0 achieved 90.74% accuracy with 90.93% precision and 90.47% recall, during testing it achieved an accuracy of 87.26%.
2. DenseNet121 achieved 80.95% training accuracy with 82.78% precision and 78.87% recall, and during testing it achieved an accuracy of 83.51%.
3. ProtoPNet achieved 93.55% training accuracy with 93.59% precision and 93.55% recall, and outperformed other models during testing with the highest accuracy of 85.67%.

References

[1] A. Iqbal et al., "Enhancing brain tumour multi-classification using EfficientNet-B0 architecture," *Information*, vol. 15, no. 8, 2024.

[2] M. J. Islam and Y. Zhang, "Brain tumor classification using MRI images and deep learning techniques," *PLOS ONE*, vol. 19, no. 5, p. e0301234, 2024.